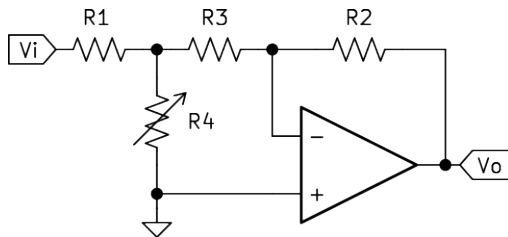


Homework 6

Due March 15th

Problems:

- Look up the datasheet for the Analog AD8531 amplifier, find the component's specified values that would fully define the dominant pole (single pole) model introduced in the class, and complete the model. Note that the "Large Signal Voltage Gain" is effectively "open-loop gain" for our purposes. Using your model, estimate the open loop gain and phase of the amplifier at 100 Hz and 100 kHz. Hint: the open loop bandwidth is commonly not provided for amplifiers.
- Calculate the frequency dependent closed-loop gain, $A(jf)$, for the Exam 1 Question 1 circuit (below); assuming $r_d = \infty$, $r_o = 0$, the resistors are valued as $R_1 = R_3 = 5\text{ k}\Omega$, $R_2 = 1\text{ M}\Omega$ and R_4 is a $10\text{ k}\Omega$ potentiometer (wiper connected to ground). The op-amp used can be described with the model provided below.
 - With R_4 set to $2.5\text{ k}\Omega$, calculate the frequency dependent closed-loop gain for this circuit, the operational bandwidth and closed-loop transition frequency. Sketch the open-loop and closed-loop gains vs. frequency and mark important points.
 - The resistors are set such that when $R_4 = 10\text{ k}\Omega$, a very large gain is achieved. What is the bandwidth at this gain?



$$a(jf) = \frac{500}{1 + jf/20\text{ Hz}} \text{ V/mV}$$

- The same op-amp as used in the question above is used to create a non-inverting amplifier circuit with a DC gain of 100.
 - What is the bandwidth of the circuit? Assemble a frequency-dependent single-pole model of the gain.
 - If the input resistance of the circuit at DC is $R_i = 100\text{ M}\Omega$, determine R_i as $f \rightarrow \infty$ and sketch R_i vs. frequency and mark important points.
- Book Problem 6.25. Use exact analysis.